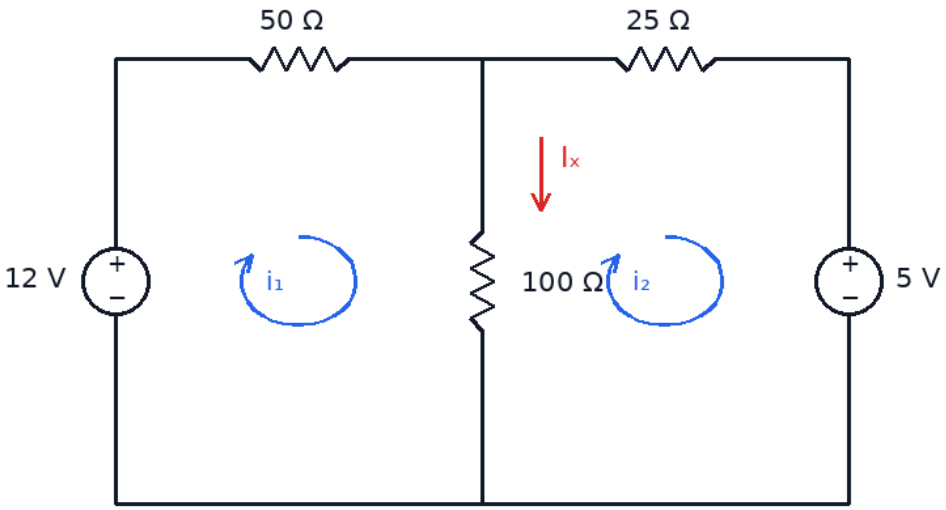
Question 1 - Homework 5

Question 1 - Mesh currents



All mesh arrows are assumed clockwise. Negative values reverse the actual direction.

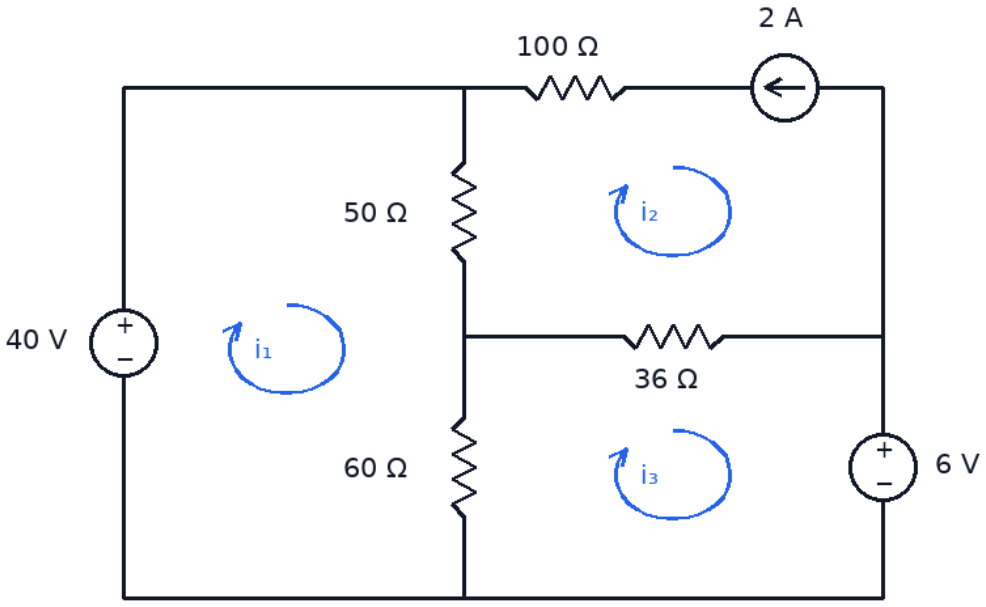
i1	i2	Ix (down)
114.286 mA	51.429 mA	62.857 mA

A) i1 = left mesh; i2 = right mesh. Both assumed clockwise.
B) Sum signed voltage drops: sum(Vdrop) = 0. A source rise is negative.
C) Left: -12 + 50i1 + 100(i1 - i2) = 0
Right: 25i2 + 5 + 100(i2 - i1) = 0
Complete set: 150i1 - 100i2 = 12; -100i1 + 125i2 = -5.
D) By hand: divide the equations by 50 and 25, respectively:
3i1 - 2i2 = 0.24; -4i1 + 5i2 = -0.20.
Multiply the first by 5 and the second by 2, then add:
7i1 = 0.80 => i1 = 4/35 A = 0.114286 A.
i2 = (3i1 - 0.24)/2 = 9/175 A = 0.051429 A.
Ix = i1 - i2 = 11/175 A = 0.062857 A (downward).
E) Passive sign convention: positive power means absorbed.
12 V: P = -12i1 = -1.371429 W, delivers 1.371429 W.
Actual current leaves its positive terminal.
5 V: P = 5i2 = +0.257143 W, absorbs 0.257143 W.
Actual current enters its positive terminal.

Currents are signed relative to the assumed clockwise mesh arrows.

Question 2 - Homework 5

Question 2 - Mesh currents



All mesh arrows are assumed clockwise. Negative values reverse the actual direction.

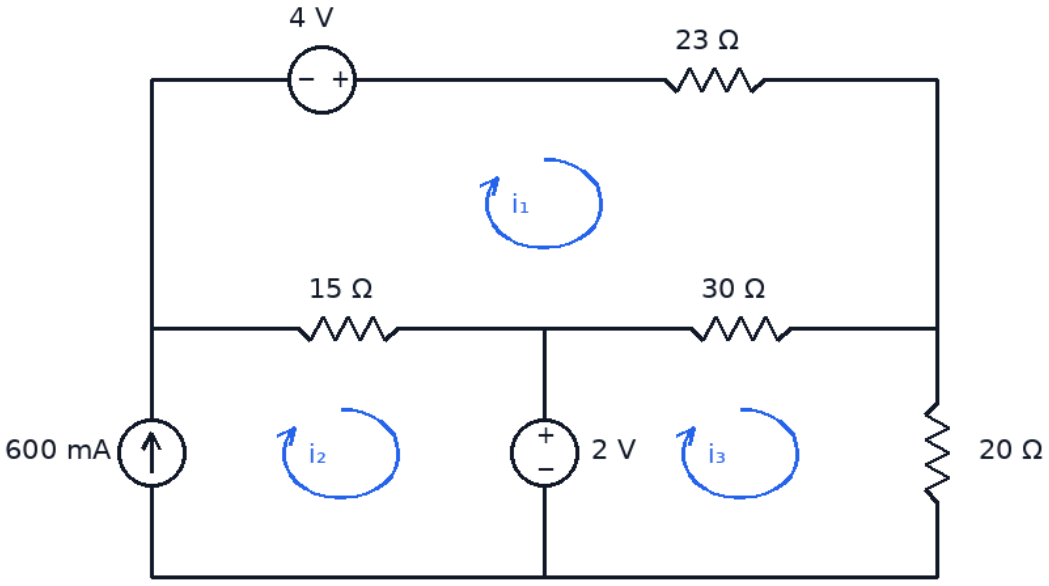
i1	i2	i3
-1.500 A	-2.000 A	-1.750 A

A) i1 = left; i2 = upper right; i3 = lower right. All clockwise.
B) Sum signed voltage drops = 0.
C) Define vc = Vleft - Vright across the 2 A source.
Left: -40 + 50(i1 - i2) + 60(i1 - i3) = 0.
Upper right: 100i2 + vc + 36(i2 - i3) + 50(i2 - i1) = 0.
Lower right: 36(i3 - i2) + 6 + 60(i3 - i1) = 0.
Constraint: i2 = -2 A (source arrow opposes clockwise).
To solve currents: 110i1 - 60i3 = -60;
-60i1 + 96i3 = -78; i2 = -2.
D) By hand: divide the first equation by 10 and the second by 6:
11i1 - 6i3 = -6; -10i1 + 16i3 = -13.
Multiply by 8 and 3, then add: 58i1 = -87 => i1 = -1.5 A.
-10(-1.5) + 16i3 = -13 => i3 = -1.75 A.
E) 40 V: P = -40i1 = +60 W, absorbs; current enters +.
6 V: P = 6i3 = -10.5 W, delivers; current leaves +.
F) Upper KVL: vc = -(186i2 - 50i1 - 36i3) = 234 V.
Source left terminal is 240 V; right terminal is 6 V.
Current enters the right (lower-voltage) terminal:
P = (6 - 240)(2) = -468 W. It delivers 468 W.

Currents are signed relative to the assumed clockwise mesh arrows.

Question 3 - Homework 5

Question 3 - Mesh currents



All mesh arrows are assumed clockwise. Negative values reverse the actual direction.

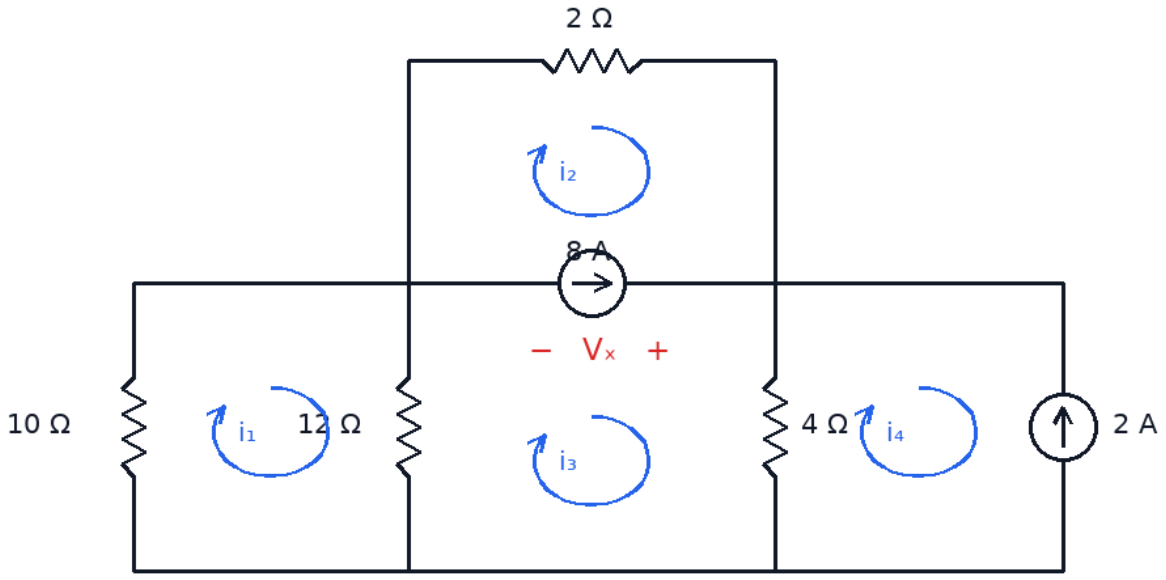
i1	i2	i3
284.0 mA	600.0 mA	210.4 mA

A) i1 = top; i2 = lower left; i3 = lower right. All clockwise.
B) Sum signed voltage drops = 0.
C) Top: -4 + 23i1 + 30(i1 - i3) + 15(i1 - i2) = 0.
Lower right: 30(i3 - i1) + 20i3 - 2 = 0.
Constraint: i2 = 0.600 A.
Complete current set: 68i1 - 15i2 - 30i3 = 4;
-30i1 + 50i3 = 2; i2 = 0.600.
If KVL is written for the source-containing lower-left loop,
define vs = Vtop - Vbottom of the 600 mA source:
15(i2 - i1) + 2 - vs = 0 (vs is an auxiliary voltage).
D) By hand: substitute i2 = 0.600:
68i1 - 30i3 = 13.
From lower-right KVL: i3 = 0.04 + 0.6i1.
68i1 - 30(0.04 + 0.6i1) = 13 => 50i1 = 14.2.
i1 = 0.284 A; i3 = 0.04 + 0.6(0.284) = 0.2104 A.
E) 4 V: current i1 leaves +; P = -4(0.284) = -1.136 W.
Delivers 1.136 W.
2 V: downward current = i2 - i3 = 0.3896 A, entering +.
P = 2(0.3896) = +0.7792 W. Absorbs 0.7792 W.

Currents are signed relative to the assumed clockwise mesh arrows.

Question 4 - Homework 5

Question 4 - Mesh currents



All mesh arrows are assumed clockwise. Negative values reverse the actual direction.

i1	i2	i3	i4	Vx
0.380952 A	-7.301587 A	0.698413 A	-2.000000 A	14.603175 V

A) i1 = left; i2 = top; i3 = center; i4 = right. All clockwise.
B) Sum signed voltage drops = 0. No supermesh is used.
C) Keep Vx = Vright - Vleft across the 8 A source.
Define v2 = Vtop - Vbottom across the 2 A source.
Left KVL: 10i1 + 12(i1 - i3) = 0.
Top KVL: 2i2 + Vx = 0.
Center KVL: -Vx + 4(i3 - i4) + 12(i3 - i1) = 0.
Right KVL: v2 + 4(i4 - i3) = 0.
Source constraints: i3 - i2 = 8; i4 = -2.
These six equations solve i1, i2, i3, i4, Vx and v2.
D) i1 = (6/11)i3; i2 = i3 - 8; Vx = -2i2 = 16 - 2i3.
Substitute into center KVL:
-(16 - 2i3) + 4(i3 + 2) + 12(i3 - 6i3/11) = 0.
(126/11)i3 = 8 => i3 = 44/63 A.
E) 8 A: current enters the negative Vx terminal.
P = -8Vx = -7360/63 = -116.825397 W; delivers.
v2 = 4(i3 - i4) = 680/63 = 10.793651 V.
2 A: upward current leaves the positive top terminal.
P = -2v2 = -1360/63 = -21.587302 W; delivers.

Currents are signed relative to the assumed clockwise mesh arrows.